

4.9 Anaphylaxis

- ▼ When are histamines and leukotrienes released?
 - When **IgE on the surface of mast cells binds to antigens.**
- ▼ What effects can histamines normally have on hosts?
 - **Peripheral vasodilation** of the arteries
 - This can lead to the appearance of a reddish rash in what is known as **urticaria**
- ▼ What is the name of the histamine receptor?
 - **GPCRs**
- ▼ When is IgE released?
 - In response to allergy and parasitic infections
- ▼ What effect does adrenaline have on blood vessels?
 - It causes **vasoconstriction.**
- ▼ What effect do histamines have on blood vessels?
 - They have the opposite effect, meaning they cause **vasodilation** and **smooth muscle contraction**
 - This can cause difficulty breathing in sufferers caused by insufficient blood supply to the heart which is seen in complications such as asthma.
- ▼ What effect do histamines have on small intestine and lungs?
 - Vasoconstriction
 - Airway constriction due to smooth muscle contraction in the airways
- ▼ What test can be used to test if someone has an allergy to something?
 - A **pin-prick test** which will usually inject small amounts of allergen into the subject to see if they produce a type of reaction known as a **wheal-and-flare reaction.**
- ▼ What two medical complications can the inhalation of allergens cause?
 - Hay fever (allergic rhinitis)
 - Asthma
- ▼ Why is this the case?
 - When an allergen is inhaled, it will act principally on the **mucosal mast cells** of the **respiratory system**

▼ What happens physiologically in hay fever?

- In hay fever, there is swelling (oedema) in the epithelia lining the nose which leads to **general irritation** and **continual running of the nose**.

▼ What happens physiologically in asthma?

- The **allergen reaches the bronchioles** through **inhalation** and causes an inflammatory response in this area
- The action of inflammatory mediators leads to the contraction of the smooth muscle in the airways
- This can lead to **breathing difficulties, seen in those with asthma**.

▼ Where does an allergen that is ingested act in the body?

- If the allergen is ingested, then it will act at mucosal mast cells in the intestinal tract.

▼ What can the ingestion of allergens lead to?

- The stimulation of the intestinal smooth muscle leads to the **vomiting and diarrhoea** typical of a food allergy.
- If the **allergic substance** can be absorbed and **pass into the blood stream** it may also cause **skin symptoms (rashes or itching)**.

▼ What does the 'shock' part of the phrase 'anaphylactic shock' specifically refer to?

- It refers to the **massive drop in blood pressure** that occurs when there is **vasodilation of the blood vessels**, which is obviously caused by the inflammatory mediators such as histamine.
- This can lead to organs not receiving the oxygen they need to function, and hence there is a risk of death

▼ What effect does salbutamol have on the body?

- It is a **bronchodilator** which can be given to people suffering from anaphylaxis, as there are breathing problems during asthma

▼ What can hypotension (low blood pressure) lead to?

- Fainting (insufficient blood flow to the brain)
- Nausea (urge to vomit)
- Vomiting

▼ What is biphasic anaphylaxis?

- A type of allergic reaction where the **symptoms disappear but then reappear after a short time.**
- **Patients** with moderate respiratory or cardiovascular compromise **should be observed for to up to 8-10 hours before discharge.**